

A Neutral Operator Theory for the Information Geometry of Dynamical Systems

From Philosophy to Foundations to Operations

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Abstract

We assume the empirically charitable stance that a dynamical representation is *faithful* if it reproduces one-step transport. This requires treating geometry (embeddings) and probability (kernels) as *co-primitive*, with the joint edge law $\Gamma = \nu \otimes K$ as the basic object.

Weak axioms reduce admissible losses through Γ to Csiszár f -divergences, and minimizing this *information action* yields a chart-invariant, compositional principle. In Gaussian limits, constructivist *least-information action* aligns with least physical action.

Information geometry equips smooth f with the Fisher–Rao metric. The dual affine structure establishes a reflective equilibrium between abstract model and empirical edge laws. While in machine learning the ‘natural gradient’ is canonical only in the parameter-epistemic sense, in our framework Fisher geometry arises from edge laws themselves. This allows us to reinterpret ‘naturalness’ not as an optimization convenience, but as the unique empirical metric that reconciles law and observation.

Vanishing edge divergence under the dual connections is then equivalent to measure conjugacy between idealized and observed dynamics, yielding generalized embedding guarantees. We build on this dual foundation and extend it to finite data. Diagnostic exponents, locality $\eta = 2s$ (smoothness) and learning $\zeta = 2s/(2s + d)$ (dimension), tie smoothness and dimension to convergence rates, while operational tests lend confidence to whether an empirical representation is predictive and faithful to a hypothesized dynamic.

1 Philosophy

Primitive object. The empirical unit is (ν, K, D) via $\Gamma = \nu \otimes K$ and a divergence.

Least-information action (LIA). Fidelity is quantified by edge divergence between data and model edge laws, optimized over representation and kernel.

2 Foundations

We fix notation and state minimal assumptions before developing the divergence-based geometry and its consequences.

2.1 Standing assumptions (minimal)

Assumption 1 (Stationarity). *The source process $(X_t)_{t \in \mathbb{Z}}$ is a time-homogeneous stochastic process on a measurable state space $(X, \mathcal{B})$ admitting an invariant probability measure μ . That is, for all*

measurable $A \in \mathcal{B}$ and all $t \in \mathbb{Z}$,

$$\mathbb{P}(X_t \in A) = \mu(A).$$

Assumption 2 (Regularization of divergences). *We assume that joint edge laws admit meaningful comparison under the chosen divergence. Concretely, this may be ensured by mutual absolute continuity with respect to a common dominating measure, by working on a finite partition, or by introducing a small noise floor to avoid singular supports. These conditions guarantee that $D_f(\Gamma_1 \parallel \Gamma_2) = 0$ implies $\Gamma_1 = \Gamma_2$ almost everywhere, avoiding pathological singularities.*

Assumption 3 (Ideal representation). *We posit the existence of an idealized process (X_t) with joint edge law Γ on $X \times X$, which serves as a conceptual reference. The observed process arises through a measurable observation map $\Psi : X \rightarrow Y$, producing an empirical kernel K and edge law Γ_K on Y . This assumption reflects our empirical stance: we do not claim the ideal representation is directly accessible, but use it as a standard against which faithfulness can be assessed.*

2.2 Objects

Let $K(y_0, dy_1)$ be a Markov kernel on an observation space Y .

Definition 1 (Source law). *The source law ν is the marginal distribution of the initial state Y_0 . It represents the empirical frequency with which states appear as origins of transport, and provides the weighting under which the kernel K acts.*

Definition 2 (Joint edge law). *Let (Y_0, Y_1) denote consecutive states of the observed process. Given a source law ν for Y_0 and a transition kernel $K : \mathcal{Y} \times \mathcal{B}(\mathcal{Y}) \rightarrow [0, 1]$, the joint edge law Γ_K is the probability measure on $\mathcal{Y} \times \mathcal{Y}$ defined by*

$$\Gamma_K(A \times B) = \int_A K(y_0, B) \nu(dy_0), \quad A, B \in \mathcal{B}(\mathcal{Y}).$$

Equivalently, Γ_K admits the disintegration

$$\Gamma_K(dy_0, dy_1) = \nu(dy_0) K(y_0, dy_1),$$

so that $Y_0 \sim \nu$ and, conditional on $Y_0 = y_0$, we have $Y_1 \sim K(y_0, \cdot)$. Thus Γ_K describes the law of a single transition, or “edge”, of the process.

Definition 3 (Edge divergence). *Let D_f be a Csiszár f -divergence between probability measures. Given two transition kernels K_1, K_2 on a common state space with common source law ν , their edge divergence is*

$$D_f^\nu(K_1, K_2) := D_f(\Gamma_{K_1} \parallel \Gamma_{K_2}),$$

where $\Gamma_{K_i} = \nu(dy_0) K_i(y_0, dy_1)$. In other words, the divergence between kernels is evaluated by lifting them to their induced joint edge laws.

Definition 4 (Primitive empirical object). *The primitive empirical object is the triple (ν, K, D_f) : a source law ν , a one-step kernel K , and an f -divergence D_f acting on the joint edge law Γ_K . This triple forms the unit of analysis for empirical representations of dynamics.*

By working with (ν, K, D_f) , we treat geometry (embeddings via ν) and probability (kernels K) as co-primitive, with divergence D_f providing the variational principle that links them.

Definition 5 (Faithfulness under an observation map). *Given a measurable observation map $\Psi : X \rightarrow Y$, define the transported edge law*

$$\Psi^\sharp(\Gamma) := (\Psi \times \Psi)^\sharp \Gamma,$$

where $(\cdot)^\sharp$ abstracts the choice of pushforward or pullback. An empirical edge law Γ_K is faithful via Ψ if

$$D_f(\Gamma_K, \Psi^\sharp(\Gamma)) = 0.$$

Faithfulness thus requires that, under Ψ , the empirical edge law coincides with the transported ideal law.

2.3 Irreducibility (no-reduction) lemmata

We first note that neither geometry alone nor probability alone suffices to decide empirical equivalence.

Lemma 1 (Geometry-only insufficient). *There exist systems with identical smooth maps $\psi : Y \rightarrow Y$ but distinct kernels, e.g. deterministic $K^{(1)}(y, \cdot) = \delta_{\psi(y)}$ and noisy $K^{(2)}(y, \cdot) = \mathcal{N}(\psi(y), \sigma^2 I)$, having different edge laws. Hence geometry alone cannot decide empirical equivalence of one-step transport.*

Lemma 2 (Probability-only insufficient). *There exist measure-isomorphic systems that are not topologically or smoothly conjugate. Measure isomorphism preserves entropy but not continuity or embedding structure. Thus probability alone cannot settle geometric equivalence.*

2.4 Axioms for admissible empirical loss

Let $\mathcal{L}(\nu, K_1, K_2)$ be a loss used to compare dynamics. We posit admissibility conditions:

- **A1 (Coordinate invariance)**: For bijective measurable h , $\mathcal{L}(\nu, K_1, K_2) = \mathcal{L}(h_\# \nu, h_\# K_1, h_\# K_2)$.
- **A2 (Data processing)**: For any Markov M , $\mathcal{L}(\nu, K_1 M, K_2 M) \leq \mathcal{L}(\nu, K_1, K_2)$ and $\mathcal{L}(M_\# \nu, M K_1, M K_2) \leq \mathcal{L}(\nu, K_1, K_2)$.
- **A3 (Temporal additivity)**: For independent blocks, $\mathcal{L}$ adds over products.
- **A4 (Locality)**: Small changes in joint edge laws yield small changes in $\mathcal{L}$.

Proposition 1 (Axioms force divergence on edge laws). *Under A1–A4, $\mathcal{L}$ factors through the joint edge laws and is (up to equivalence) an f -divergence:*

$$\mathcal{L}(\nu, K_1, K_2) = D_f(\Gamma_{K_1} \parallel \Gamma_{K_2}).$$

2.5 Local geometry

If K_θ is a smooth family and f is smooth with $f''(1) > 0$, then

$$D_f^\nu(K_{\theta+\delta\theta}, K_\theta) = \frac{1}{2} \delta\theta^\top I_f(\theta) \delta\theta + o(\|\delta\theta\|^2),$$

which defines the Fisher–Rao (information) metric I_f on the parameter manifold Θ . This is the intrinsic, reparameterization–invariant geometry relevant for local inference.

State-space pullback. Viewing $y \mapsto K(\cdot|y)$ as a map into a statistical manifold, pulling back Fisher yields

$$G(y) := \mathbb{E}_{K(\cdot|y)}[\nabla_y \log k(Y_1|y) \nabla_y \log k(Y_1|y)^\top],$$

a tensor field that measures how distinguishable outgoing conditionals are under small moves in y .

In practice we use the stable mean-only proxy

$$J(y) \approx (\partial_y \mu(y))^\top \Sigma(y)^{-1} (\partial_y \mu(y)),$$

where $\mu(y) = \mathbb{E}[Y_1|y]$ and $\Sigma(y) = \text{Var}[Y_1|y]$. Basis/lag selection is then posed as maximizing Fisher capture $\text{trace}(L^\top \bar{J} L)$ or $\log \det(L^\top \bar{J} L)$, with $\bar{J} := \mathbb{E}_\nu[J(Y)]$.

Gaussian local model. If $K(y, \cdot) \approx \mathcal{N}(\mu(y), \Sigma(y))$ with smooth μ, Σ , neglecting $\partial_y \Sigma$ gives

$$J(y) \approx (\partial_y \mu(y))^\top \Sigma(y)^{-1} (\partial_y \mu(y)),$$

the familiar Gauss–Newton/Fisher form. In a local linear fit $Y_{t+1} \approx A(y_t)Y_t + b(y_t)$, we set $\partial_y \mu(y) \approx A(y)$ so $J(y) \approx A(y)^\top \Sigma(y)^{-1} A(y)$.

Remark 1 (Minimality of the triple (ν, K, D_f)). *Remove $K \Rightarrow$ no transport/prediction; remove $D_f \Rightarrow$ no ruler/decision; remove $\nu \Rightarrow$ no source weighting. Add path laws and additivity collapses them back to edge laws. Hence (ν, K, D_f) is both sufficient and minimal for empirical comparison.*

Remark 2 (Why one-step is primitive). *Pathwise divergences (e.g. KL) decompose into sums of edge-level terms for Markov/i.i.d. blocks; thus the edge divergence is the atomic unit, with path “action” an additive aggregate.*

2.6 From local to global

Lemma 3 (Zero divergence $\Rightarrow$ kernel equality). *If $D_f(\Gamma_{K_1} \|\Gamma_{K_2}) = 0$ and $\Gamma_{K_1} \ll \Gamma_{K_2}$ (or conversely), then $K_1(y_0, \cdot) = K_2(y_0, \cdot)$ for ν -a.e. y_0 .*

Theorem 1 (Sufficiency $\Rightarrow$ Fisher equality). *If Ψ is sufficient for Γ , then for any smooth Csiszár f -divergence with $f''(1) > 0$, the local quadratic form is preserved:*

$$I_\Gamma(\theta) = I_{\Psi(\Gamma)}(\theta).$$

Remark 3 (Local vs global). *Sufficiency preserves Fisher information (a local, second-order object), but does not by itself imply equality of edge laws. Global conjugacy requires vanishing divergence, a stronger condition we develop below.*

Theorem 2 (Generalized embedding theorem). *Let $h = \Psi_k$ be an embedding map with $\nu = h_\# \mu$. With K_X the kernel on X and K_Y on Y ,*

$$D_f''(K_Y, h_\# K_X) = 0 \iff K_X \text{ and } K_Y \text{ are measure-conjugate via } h.$$

Sketch. Zero divergence and domination imply $K_Y(y, \cdot) = h_\# K_X(y, \cdot)$ for ν -a.e. y (Lemma 3). Iterating h gives equality of n -step cylinder measures, hence measure conjugacy. $\square$

Corollary 1 (Faithfulness $\Rightarrow$ measure conjugacy). *If Γ_K is faithful to Γ via a diffeomorphic Ψ , i.e. $D_f(\Gamma_K, \Psi(\Gamma)) = 0$, then Γ_K and Γ are measure-conjugate through Ψ .*

2.7 Dual affine geometry of divergence

The Fisher metric g equips the statistical manifold of kernels with a Riemannian structure, but curvature alone does not explain why divergences decompose additively or why certain parameterizations are privileged. A further refinement is provided by *dual affine connections*, which pair with the Fisher metric to yield a dually flat structure.

Definition 6 (Dual connections). *Let $(\mathcal{M}, g)$ be a statistical manifold with Fisher metric g . Two affine connections ∇ and ∇^* are said to be dual with respect to g if for all vector fields X, Y, Z ,*

$$X[g(Y, Z)] = g(\nabla_X Y, Z) + g(Y, \nabla_X^* Z).$$

Amari's α -connections $\{\nabla^{(\alpha)} : \alpha \in \mathbb{R}\}$ form a one-parameter family of mutually dual connections. Of particular importance are $\alpha = 1$ (the exponential or e -connection) and $\alpha = -1$ (the mixture or m -connection).

Remark 4 (Exponential and mixture flatness). *Exponential families are affine under $\nabla^{(e)}$, so their natural parameters form an e -flat coordinate system. Mixture families are affine under $\nabla^{(m)}$, so their mixture weights form an m -flat coordinate system. The two are Legendre dual via the log-partition function.*

Definition 7 (Amari projections). *Given a distribution p and a model family $\mathcal{M}$,*

$$\begin{aligned} m\text{-projection: } q^* &= \arg \min_{q \in \mathcal{M}} D_{\text{KL}}(p \| q), \\ e\text{-projection: } q^* &= \arg \min_{q \in \mathcal{M}} D_{\text{KL}}(q \| p). \end{aligned}$$

The m -projection is orthogonal with respect to the m -connection, while the e -projection is orthogonal with respect to the e -connection.

Theorem 3 (Pythagorean theorem for KL divergence). *If $\mathcal{M}$ is e -flat and $\mathcal{N}$ is m -flat, then for $p \in \mathcal{N}$ and q^* the m -projection of p onto $\mathcal{M}$,*

$$D_{\text{KL}}(p \| q) = D_{\text{KL}}(p \| q^*) + D_{\text{KL}}(q^* \| q), \quad \forall q \in \mathcal{M}.$$

Thus KL divergence decomposes orthogonally into increments along the dual affine flats.

Corollary 2 (Edge laws as dual projections). *Let Γ_{data} denote the empirical edge law, and let $\mathcal{M} = \{\Gamma_K : K \in \mathcal{K}\}$ be an e -flat family of candidate kernels. Then the m -projection of Γ_{data} onto $\mathcal{M}$ is the model kernel Γ_{K^*} minimizing the divergence,*

$$\Gamma_{K^*} = \arg \min_{\Gamma_K \in \mathcal{M}} D_{\text{KL}}(\Gamma_{\text{data}} \| \Gamma_K).$$

For any $\Gamma_K \in \mathcal{M}$,

$$D_{\text{KL}}(\Gamma_{\text{data}} \| \Gamma_K) = D_{\text{KL}}(\Gamma_{\text{data}} \| \Gamma_{K^*}) + D_{\text{KL}}(\Gamma_{K^*} \| \Gamma_K).$$

Thus the excess divergence of any model candidate decomposes into the irreducible divergence between data and its m -projection, plus the divergence between this projection and the candidate.

Corollary 3 (General f -divergences). *For a smooth f -divergence with $f''(1) > 0$, the exact Pythagorean relation does not hold, but the local quadratic approximation is preserved. For q^* the m -projection of p onto $\mathcal{M}$,*

$$D_f(p \| q) = D_f(p \| q^*) + \frac{1}{2} \delta \theta^\top I_f(\theta^*) \delta \theta + o(\|\delta \theta\|^2),$$

where $\delta\theta$ is the displacement from q^* to q in parameter space and I_f is the Fisher information metric induced by D_f . Thus the decomposition holds up to second order, with increments governed by Fisher geometry. In particular, the quadratic term controls the growth rates of local divergence, and hence directly determines the locality and learning exponents (η, ζ) introduced earlier.

Proposition 2 (Rose condition as intersection of dual flats). *Empirical edge laws, constructed as mixtures of observed conditionals, live on an m -flat submanifold. Candidate kernels, parameterized as local exponential families, live on an e -flat submanifold. The Rose condition*

$$D_f(\Gamma_{data}, \Gamma_K) = 0$$

is equivalent to Γ_{data} lying in the intersection of the m -flat and e -flat submanifolds. In this case, the m -projection coincides with the model itself, and the divergence vanishes.

2.8 Sufficient statistics as observation functions

Chentsov's theorem identifies the Fisher metric as the unique (up to rescaling) Riemannian metric monotone under Markov morphisms (statistics). Amari's α -connections $\{\nabla^{(\alpha)}\}$ preserve this structure, and when a statistic is sufficient the induced Fisher metric and dual connections are isometric to the originals. Thus sufficiency is not a lossy compression but a geometry-preserving reduction.

Conditional families as statistical manifolds. In our setting the one-step conditionals

$$\mathcal{F} = \{K(\cdot | y) : y \in Y\}$$

form a statistical manifold, with the state y acting as a parameter. If these conditionals are locally exponential families with sufficient statistic T , then the expectation parameter

$$m(y) := \mathbb{E}[T(Y_1) | Y_0 = y]$$

provides an m -affine coordinate system on $\mathcal{F}$.

Lemma 4 (Expectation map as embedding). *Suppose $m : A \rightarrow \mathbb{R}^k$ is C^1 on the attractor $A \subseteq Y$, with $\text{rank } \partial m(y) = \dim A$ for all $y \in A$ (immersion), and m is injective on A . Then $F(y) = m(y)$ is a diffeomorphic embedding of A onto its image. In this case, the sufficient statistic T both preserves the α -geometry of the conditional family and yields an observation function that separates states.*

Sketch. Immersion + injectivity ensures m provides smooth coordinates on the attractor, hence a diffeomorphic embedding. $\square$

Corollary 4 (Gaussian case). *If $K(\cdot | y) \approx \mathcal{N}(\mu(y), \Sigma)$ with fixed Σ , then $T(y_1) = (y_1, y_1 y_1^\top)$ is sufficient and $m(y) = (\mu(y), \mu(y)\mu(y)^\top + \Sigma)$. If $\mu : Y \rightarrow \mathbb{R}^d$ is an injective immersion on A , then $F(y) = \mu(y)$ itself is an embedding. This coincides with the familiar local Fisher picture, where $J(y) \approx (\partial_y \mu(y))^\top \Sigma^{-1} (\partial_y \mu(y))$.*

Thus sufficiency on the model manifold and embedding on the state manifold can coincide: the same statistic that preserves Fisher geometry of the conditional family can simultaneously serve as a Takens-style observation map. The speed at which such embeddings and sufficiency conditions become empirically detectable is governed by divergence decay rates, linking them directly to ergodic exponents and the finite-data diagnostics (η, ζ) introduced next.

2.9 Rates of divergence and dynamical critical exponents

The asymptotic decay of divergences under e- and m-projections reflects the same scaling laws that ergodic theory encodes in its critical exponents. In the limit $n, t \rightarrow \infty$, both projections converge to zero if the empirical law Γ_K is faithful to the embedded truth $\Psi(\Gamma)$, but the *rates* of convergence reveal which features of the dynamics are bulk-driven (typical set) and which are tail-driven (rare events). Here the slopes (ζ, η) operationalize these rates: η reflects local Fisher curvature, while ζ interpolates smoothness and effective dimension.

Dictionary.

Exponent	Dynamics meaning	Divergence signal
Entropy h	Orbit complexity growth	Rate of per-step KL (cross-entropy)
Lyapunov λ	Stretching of trajectories	Tail sensitivity, rare-event amplification
Recurrence ρ	Return-time scaling	Limits e-projection convergence (tails)
Dimension d_μ	Mass scaling/fractality	Gap between e- and m-projection rates
Mixing κ	Decay of correlations	Divergence contraction $\sim t^{-\kappa}$
Spectral β	Transfer/Koopman decay	Exponential decay via spectral gap

Bridge theorem (informal). Classical ergodic exponents (entropy, Lyapunov, recurrence, dimension, mixing, spectrum) manifest directly as rates at which divergences vanish. Proof sketches and detailed statements are collected in the appendix.

3 Finite data: convergence & irreducible randomness

Two asymptotic lenses (empirical diagnostics). Let $\hat{\mathcal{A}}$ denote the estimated edge action on held-out data.

- **Learning curve (sample size):** with model capacity tuned by CV at each size n ,

$$\hat{\mathcal{A}}(n) \approx \mathcal{A}_\infty + C_{\text{learn}} n_{\text{eff}}^{-\zeta}, \quad n_{\text{eff}} := n/\tau_{\text{mix}}.$$

- **Locality curve (bandwidth):** for large n , with a local estimator of scale h ,

$$\hat{\mathcal{A}}(h) \approx \mathcal{A}_0 + C_{\text{loc}} h^\eta.$$

Here τ_{mix} is an integrated autocorrelation time (effective sample-size correction). The asymptotes $\mathcal{A}_\infty$ and $\mathcal{A}_0$ coincide (up to measurement-noise corrections) in well-specified settings.

Proposition 3 (Bias–variance law for excess edge action). *For a Markov system with Hölder smooth conditional of order $s > 0$, a local polynomial estimator of order matching s , embedding dimension d , and effective size n_{eff} , the excess action admits the decomposition*

$$\mathbb{E}[\hat{\mathcal{A}}(h, n)] - \mathcal{A}_0 \approx C_b h^{2s} + C_v \frac{1}{n_{\text{eff}} h^d},$$

where C_b (curvature/Fisher-weighted) and C_v (noise/mixing-dependent) are positive constants.

Corollary 5 (Linking exponents to smoothness and dimension). *If variance is negligible (large n), then $\hat{\mathcal{A}}(h) - \mathcal{A}_0 \propto h^\eta$ with*

$$\boxed{\eta = 2s}.$$

Balancing bias and variance gives the optimal scale $h^* \asymp n_{\text{eff}}^{-1/(2s+d)}$ and the learning rate

$$\zeta = \frac{2s}{2s+d}.$$

Thus η reveals smoothness $s \approx \eta/2$, and with ζ one can back out an effective dimension

$$d \approx \frac{2s(1-\zeta)}{\zeta}.$$

Remark 5 (Stochastic floors and irreducible randomness). *If (after correcting for measurement noise) both asymptotes satisfy $\mathcal{A}_\infty \approx \mathcal{A}_0 > 0$, then no further geometric refinement (embedding), capacity growth, or locality contraction will collapse the action: this is a signature of irreducible process noise. If either asymptote tends to 0, the apparent randomness was due to aliasing, underfitting, or finite data.*

4 Pathwise and deterministic limits (how least physical action emerges)

We recover least physical action as a continuum limit of summed edge actions under forward-KL and Gaussian local models.

Proposition 4 (Gaussian forward-KL $\Rightarrow$ quadratic path action). *Let time be discretized with step Δt . Suppose the model kernel is locally Gaussian:*

$$K_\theta(y, dy') \approx \mathcal{N}(y + \mu_\theta(y) \Delta t, \Sigma(y) \Delta t).$$

Then the summed edge action $\sum_t D_{\text{KL}}(\Gamma_{\text{data}}^{(t)} \parallel \Gamma_{K_\theta}^{(t)})$ Riemann-sums to a quadratic path functional

$$\mathcal{A}_{\text{path}}(\theta) = \frac{1}{2} \mathbb{E} \int_0^T \|\dot{y}_t - \mu_\theta(y_t)\|_{\Sigma(y_t)^{-1}}^2 dt + \text{const.}$$

Remark 6 (Onsager–Machlup/Girsanov bridge). *For diffusions $dY_t = b(Y_t) dt + \sigma dW_t$, changing drift to $b + u$ yields a path-space KL equal to $\frac{1}{2} \mathbb{E} \int_0^T \|u_t\|_{(\sigma\sigma^\top)^{-1}}^2 dt$. Thus most-probable paths under small noise extremize the quadratic stochastic action, aligning LIA with Onsager–Machlup.*

Proposition 5 (Small-noise/continuum limit $\Rightarrow$ least physical action). *In the limit of vanishing noise and $\Delta t \rightarrow 0$, stationary paths of $\mathcal{A}_{\text{path}}$ satisfy Euler–Lagrange/Hamilton–Jacobi equations associated with the induced quadratic Lagrangian. Hence classical least action is recovered as a limit of LIA.*

5 Operations (how it collapses practice)

This section collapses the theory into a single scalar diagnostic (edge action) with two slopes (ζ, η) and an information-guided selector.

One scalar: edge action. Fix a representation and divergence; estimate on a held-out block the edge action

$$\mathcal{A} := \widehat{D}_f^\nu(K_{\text{data}}, K_{\text{model}}),$$

with a block-bootstrap confidence interval. Decision: CI contains 0 (and mean tiny) $\Rightarrow$ empirical Rose pass; with Takens, invoke Theorem 2. Else, adjust basis or kernel.

Optional e -side diagnostic (capacity sweep). To expose the complementary e -side contraction, construct a capacity/regularization sweep for the model family $\{\Gamma_\theta\}$ and track the e -excess $D_{\text{KL}}(\Gamma_\theta \parallel \Gamma_{\text{data}}) - D_{\text{KL}}(\Gamma_{\theta^*} \parallel \Gamma_{\text{data}})$. On log-log axes (excess vs. effective capacity or sample size), the slope estimates the e -side learning exponent ζ . In practice this uses the same kernels/embeddings as the m -side, and shares hyperparameter infrastructure (CV/bandwidth grids), so it adds a single loop without altering the main workflow.

Two robust implementations (pick one).

- **VQ-JS (discrete, stable):** global codebook on y_1 (k-means, M cells). For each held-out y_0 : empirical conditional (kNN over edges) $\rightarrow$ histogram p (smoothed); model conditional (sample from local Gaussian kernel) $\rightarrow$ histogram q (same cells, smoothed); compute $\text{JS}(p, q)$ and average across y_0 .
- **Parzen-KL (continuous, MC):** smooth empirical conditional $\hat{p}_b(y_1|y_0)$ (Parzen); approximate $D_{\text{KL}}(\hat{p}_b \parallel \hat{K})$ by Monte Carlo; average across y_0 .

Kernel estimator (model). Local linear mean via weighted least squares $Y_{t+1} \approx A(y_0)Y_t + b(y_0)$, residual diffusion

$$\Sigma(y_0) = \frac{\sum w_t r_t r_t^\top}{\sum w_t} + \lambda I, \quad r_t = Y_{t+1} - A(y_0)Y_t - b(y_0),$$

and $\hat{K}(y_0, \cdot) = \mathcal{N}(\mu(y_0), \Sigma(y_0))$ (mixtures as needed). Tune neighborhood scale h , ridge λ (and mixture size) by held-out conditional log-likelihood.

Information-guided basis and lag selection. Aggregate the pullback metric by visitation to obtain a global selector

$$\bar{J} := \mathbb{E}_\nu[J(Y)] \approx \frac{\sum_t w_t A_t^\top \Sigma_t^{-1} A_t}{\sum_t w_t},$$

using the same locality weights w_t as in the kernel fit.

Basis (rotation). For a linear projection $L \in \mathbb{R}^{d \times r}$ with $L^\top L = I_r$, maximize the *Fisher capture* $\text{trace}(L^\top \bar{J} L)$. The optimizer is given by the top- r eigenvectors of $\bar{J}$ (information-PCA). This choice is invariant to invertible linear reparameterizations.

Lag (subset) selection. When y is a delay stack, $\bar{J}$ is block-structured by lags. Two principled criteria are:

- **A-optimal (trace):** choose S to maximize $\text{trace}(\bar{J}_{SS})$;
- **D-optimal (log-det):** choose S to maximize $\log \det(\bar{J}_{SS})$ (encourages non-redundant lags).

A stable greedy D-optimal procedure:

1. Initialize $S \leftarrow \emptyset$.
2. For each candidate $\ell \notin S$, compute the Schur-complement gain $\Delta_\ell = \log \det(J_{\ell\ell} - J_{\ell S} J_{SS}^{-1} J_{S\ell})$.
3. Add $\ell^* = \arg \max_\ell \Delta_\ell$ to S and iterate until Δ_{ℓ^*} is below a tolerance or a CV-selected budget is reached.

Refit & certify. After selecting L (and/or S), refit the kernel in the new representation and recompute (i) the edge action with a block-bootstrap CI and (ii) the learning/locality exponents (ζ, η) . A better basis typically lowers the edge action (or tightens its CI), increases ζ , and steepens η .

Notes. Pre-whiten y for stability; use a small ridge on Σ_t . If $\Sigma(y)$ varies strongly, include the covariance-variation contribution to $J(y)$; otherwise the mean term dominates in practice.

Outputs. Report $\mathcal{A}$ with CI (primary), predictive NLL (sanity), and optionally moment gaps / quick spectral snapshot.

6 Checklist (logical closure)

- **Irreducibility:** Geometry or probability alone is insufficient (Lemmas 1, 2).
- **Axioms $\Rightarrow$ f -divergences:** Proposition 1.
- **Zero-divergence lemma:** Equality-from-zero (Lemma 3).
- **Takens+embedding:** With a delay embedding, vanishing edge divergence implies measure conjugacy (Theorem 2).
- **Dual-affine geometry:** Projections and Pythagorean decompositions explain additive divergence structure.
- **Finite-data diagnostics:** Exponents (η, ζ) expose smoothness and dimension, and operational tests certify faithfulness.
- **Continuum link:** Gaussian/forward-KL limits recover least physical action.

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A Divergences and Critical Exponents

This section collects the technical lemmas linking divergence decay rates to classical dynamical critical exponents.

Lemma 5 (Contraction under transfer operator). *Let P be the Perron–Frobenius operator with invariant measure ν . For any f -divergence,*

$$D_f(P\mu\|P\nu) \leq \eta_f(P) D_f(\mu\|\nu),$$

where $\eta_f(P) \in [0, 1]$ is a contraction coefficient. If P has a spectral gap with decay $e^{-\gamma t}$, then $D_f(P^t\mu\|P^t\nu)$ decays at rate γ , corresponding to the system’s mixing/spectral exponent.

Lemma 6 (Entropy rate). *For stationary processes,*

$$\frac{1}{n} D_{\text{KL}}(\Gamma^{\otimes n} \| \Gamma_K^{\otimes n}) \rightarrow H(\Gamma \| \Gamma_K),$$

the cross-entropy rate. If $\Gamma = \Gamma_K$, the limit vanishes and the slope constants are controlled by the entropy rate h .

Lemma 7 (Recurrence tails). *If return times τ_ε satisfy $\mathbb{P}(\tau_\varepsilon > t) \sim t^{-\rho}$, then under-coverage of rare returns limits the rate of e -projection convergence by ρ .*

Lemma 8 (Dimension exponent). *Locality and learning curves obey*

$$\hat{A}(h, n) - A_0 \asymp C_b h^{2s} + \frac{C_v}{n_{\text{eff}} h^{d_\mu}},$$

so the effective dimension d_μ appears as the scaling exponent in divergence decay. The gap between e - and m -projections is larger when d_μ is small (singular measures).

Theorem 4 (Bridge between divergences and exponents). *Let P be the transfer operator of a stationary dynamical system with invariant measure ν , admitting a spectral gap with decay exponent γ . Suppose Ψ is a diffeomorphic delay embedding. Then for any smooth f -divergence,*

$$D_f(P^t \Gamma_K \| P^t \Psi(\Gamma)) \vee D_f(P^t \Psi(\Gamma) \| P^t \Gamma_K) \leq C e^{-\gamma t} + R_n,$$

where R_n is the finite-sample estimation error scaling with smoothness s , dimension d_μ , and recurrence exponent ρ . In particular, as $n, t \rightarrow \infty$, both projections vanish, and the relative convergence rates diagnose whether the dynamics are tail-driven (slower e) or bulk-driven (slower m).

Remark. These results establish the dictionary promised in the main text.

B Computational Routine for LIA

This appendix provides a step-by-step description of how the least-information action (LIA) is evaluated in practice when the edge divergence is chosen as the Maximum Mean Discrepancy (MMD). The routine collapses the theoretical framework to an implementable sequence of computations.

B.1 Primitive Setup

1. **Inputs.** A time series $\{y_t\}_{t=0}^n$, a kernel k (RBF or Fisher), kernel scale ℓ , and a candidate representation (embedding h , neighborhood metric/bandwidths).
2. **Objective.** Estimate the *edge action*

$$A = \hat{D}_{\text{MMD}}^\nu(K_{\text{data}}, K_{\text{model}}) \approx \frac{1}{|\mathcal{Q}|} \sum_{Y_0 \in \mathcal{Q}} \text{MMD}^2(\hat{K}_{\text{data}}(\cdot | Y_0), \hat{K}_{\text{model}}(\cdot | Y_0)),$$

where $\mathcal{Q}$ is a set of query points drawn from the test block, and report a block-bootstrap confidence interval (CI). If the CI contains 0 and the mean is small, the empirical Rose condition is considered satisfied.

B.2 Local Mean Fit (for representation and geometry)

1. **Local linear mean.** For each query point $Y_0 \in \mathcal{Q}$, estimate

$$Y_{t+1} \approx A(Y_0)Y_t + b(Y_0),$$

using weighted least squares with exponential weights

$$w_t = \exp\left(-\frac{\|\tilde{Y}_t - \tilde{Y}_0\|_2}{h}\right),$$

where each coordinate of Y is z-scored on the training block.

2. **Residuals (for bootstrap).** Define $r_t = y_{t+1} - A(Y_0)^\top Y_t - b(Y_0)$; the weighted empirical residual law $\hat{R}_{Y_0} \propto \sum_t w_t \delta_{r_t}$ will be used to generate model-side fluctuations nonparametrically.

B.3 Divergence Estimation (MMD)

For each $Y_0 \in \mathcal{Q}$, construct both empirical and model conditionals:

1. **Empirical conditional.** Using the test block, form weights $\omega_t(Y_0) \propto \exp(-\|\tilde{Y}_t - \tilde{Y}_0\|_2/h_p)$ and define $\hat{K}_{\text{data}}(\cdot|Y_0) = \sum_t \pi_t \delta_{y_{t+1}}$, $\pi_t = \omega_t / \sum_{t'} \omega_{t'}$.
2. **Model conditional (pushforward + residual bootstrap).** Draw $\epsilon \sim \hat{R}_{Y_0}$ and set $\tilde{y}_{t+1} = A(Y_0)^\top Y_0 + b(Y_0) + \epsilon$, repeating L times to obtain bootstrap samples from $\hat{K}_{\text{model}}(\cdot|Y_0)$.
3. **MMD estimate.** With kernel $k_\ell(y, y')$, compute

$$\text{MMD}^2(\hat{P}_{Y_0}, \hat{Q}_{Y_0}) = \sum_{t, t'} \pi_t \pi_{t'} k_\ell(y_{t+1}, y_{t'+1}) + \frac{1}{L^2} \sum_{j, j'} k_\ell(\tilde{y}_j, \tilde{y}_{j'}) - \frac{2}{L} \sum_j \sum_t \pi_t k_\ell(\tilde{y}_j, y_{t+1}).$$

4. **Aggregate.** Average over $Y_0 \in \mathcal{Q}$ to obtain $\hat{A}$.

Kernel choices.

- **RBF (default).** $k_\ell(y, y') = \exp(-\frac{(y-y')^2}{2\ell^2})$, with ℓ chosen by median heuristic or small validation sweep.
- **Fisher kernel (local FR geometry).** For local Gaussian mean model at Y_0 ,

$$k_{Y_0}(y, y') = \frac{(y - \hat{\mu}(Y_0))(y' - \hat{\mu}(Y_0))}{\hat{\sigma}(Y_0)^2},$$

aligning MMD's quadratic form with the Fisher–Rao metric.

B.4 Information-Guided Basis and Lag Selection

1. **Pullback Fisher proxy.** Compute

$$\bar{J} = \mathbb{E}_\nu[J(Y)] \approx \frac{\sum_t w_t A_t^\top (\widehat{\text{Var}}[r_t|Y_t])^{-1} A_t}{\sum_t w_t}.$$

2. **Basis selection.** For a linear projection $L \in \mathbb{R}^{d \times r}$ with $L^\top L = I_r$, maximize $\text{trace}(L^\top \bar{J} L)$ (information PCA).
3. **Lag selection.** Perform greedy D-optimal selection of lags by maximizing marginal gains in $\log \det(\bar{J}_{SS})$ via Schur complement.

B.5 Diagnostics

- **Learning curves.** Estimate $\hat{A}(n)$ versus n on log–log scale, fit slope ζ , and recover asymptotic error A_∞ .
- **Locality curves.** Estimate $\hat{A}(h)$ versus h on log–log scale, fit slope η , and recover limiting error A_0 .
- **Interpretation.** ζ, η identify effective smoothness s and dimension d ; $A_\infty \approx A_0 > 0$ signals an irreducible process-noise floor.

C Simulation Protocols

C.1 Logistic Map with Tunable Diffusion

Purpose. A minimal nonlinear, scalar system that demonstrates: near-zero edge action under good representation, recovery of learning/locality exponents (ζ, η) , information-guided lag selection, and the emergence of an irreducible process-noise floor as diffusion increases.

C.1.1 Data-Generating Model

Let $x_t \in [0, 1]$ evolve by

$$x_{t+1} = r x_t (1 - x_t) + \eta_t, \quad r = 3.8, \quad \eta_t \sim \mathcal{N}(0, \sigma^2).$$

Observations are $y_t := x_t$. Diffusion strength is swept over $\sigma \in \{0, 0.005, 0.01, 0.02, 0.05\}$.

C.1.2 Protocol

1. **Trajectories.** For each σ , simulate $n = 100,000$ points; discard a burn-in of 10,000.
2. **Splits.** Use contiguous 60%/20%/20% train/val/test blocks.
3. **Representation.** Delay-embed y_t into k lags: $Y_t = [y_t, y_{t-1}, \dots, y_{t-k+1}]^\top$, with $k \in \{2, \dots, 8\}$.
4. **Local mean fit.** For each Y_0 in the test block, fit $y_{t+1} \approx a(Y_0)^\top Y_t + b(Y_0)$ with exponential weights $w_t = \exp(-\|\tilde{Y}_t - \tilde{Y}_0\|_2/h)$. Record residuals r_t and the weighted empirical residual law $\hat{R}_{Y_0}$.
5. **Hyperparameters.** Tune h (and ridge) by validation conditional log-likelihood. Sweep h on a log grid from $c_{\min} \cdot \text{median}(d_t)$ to $c_{\max} \cdot \text{median}(d_t)$ with $c_{\min} = 0.25$, $c_{\max} = 8$, using $m \in \{10, 12, 16\}$ points.
6. **Edge divergence (MMD).** For each test query Y_0 :
 - (a) Empirical: build $\hat{P}_{Y_0}$ with test-block weights h_p .
 - (b) Model: bootstrap L residual samples to form $\hat{Q}_{Y_0}$.
 - (c) Compute $\text{MMD}^2(\hat{P}_{Y_0}, \hat{Q}_{Y_0})$ with kernel k_ℓ .
7. **Uncertainty.** Block bootstrap the test block with block length $B \approx 5 \tau_{\text{mix}}$ to produce a 95% CI for $\hat{A}$.

8. **Information-guided lag selection.** Estimate $\bar{J}$ on the training block. Perform greedy D-optimal lag selection over $\{1, \dots, 8\}$; refit and recompute $\hat{A}$.
9. **Learning and locality exponents.** (i) Compute $\hat{A}(n)$ for subsample sizes $n \in \{10^3, 2 \times 10^3, \dots, 10^5\}$ and fit $\hat{A}(n) \approx A_\infty + Cn_{\text{eff}}^{-\zeta}$.
(ii) Sweep h across the log grid and fit $\hat{A}(h) \approx A_0 + Ch^\eta$.

C.1.3 Expected Recoveries

- **Edge action.** Adequate embeddings yield $\hat{A} \approx 0$ with CI covering zero when $\sigma \approx 0$; misspecified embeddings yield $\hat{A} > 0$.
- **Exponents.** $\eta \approx 2s$ recovers smoothness s , while $\zeta \approx \frac{2s}{2s+d}$ identifies effective dimension d .
- **Noise floor.** As σ grows, A_∞ and A_0 coalesce above zero, revealing irreducible process noise.
- **Lag selection.** D-optimal lag choice stabilizes estimation, typically selecting $k \in \{3, 4\}$ at $r=3.8$.

C.1.4 Default Knobs

- Kernel: RBF with scale ℓ (median heuristic) or Fisher kernel.
- Locality $h \in \{0.5, 1, 2, 4\} \cdot n^{-1/5}$; decouple (h_p, h_r) via multipliers $(\gamma_{hp}, \gamma_{hr})$.
- Model samples $L = 5,000$.
- Bootstrap resamples = 500; block length $B \approx 5 \tau_{\text{mix}}$.

Algorithm 1 LIA with exponential kernel and conditional MMD

- 1: **Input:** $\{y_t\}$, lag set $\mathcal{K}$, kernel k (RBF or Fisher), resamples B
 - 2: Split into Train/Val/Test; build delay embeddings for each $k \in \mathcal{K}$
 - 3: **for** h in log-grid, query Y_0 in Train/Val **do**
 - 4: Fit $y_{t+1} \approx a(Y_0)^\top Y_t + b(Y_0)$ by WLS with w_t ; store residuals r_t
 - 5: **end for**
 - 6: Choose h by validation log-likelihood; set kernel scale ℓ
 - 7: **for** Test Y_0 **do**
 - 8: *Empirical* $\hat{P}_{Y_0}$: weights $\omega_t \propto \exp(-\|\tilde{Y}_t - \tilde{Y}_0\|_2/h_p)$ on Test
 - 9: *Model* $\hat{Q}_{Y_0}$: sample $\epsilon \sim \hat{R}_{Y_0}$; set $\tilde{y} = a(Y_0)^\top Y_0 + b(Y_0) + \epsilon$; repeat L times
 - 10: Compute $\text{MMD}^2(\hat{P}_{Y_0}, \hat{Q}_{Y_0})$ with kernel k_ℓ
 - 11: **end for**
 - 12: $\hat{A} \leftarrow$ average MMD^2 over test queries; moving-block bootstrap (B resamples) for CI
 - 13: $\bar{J} \leftarrow \frac{\sum w_t A_t^\top \widehat{\text{Var}}[r_t|Y_t]^{-1} A_t}{\sum w_t}$; D-optimal lag selection $\rightarrow S^*$; refit and recompute $\hat{A}$ with CI
 - 14: Build $\hat{A}(n)$ and $\hat{A}(h)$; fit slopes (ζ, η) and limits (A_∞, A_0)
 - 15: **Output:** $\hat{A} + \text{CI}$, S^* , (ζ, η) , A_∞, A_0 , $\bar{J}$ spectrum and D-optimal gains
-

C.2 Numerical Demonstration Plan

C.2.1 Guiding Principle

The theoretical results (Proposition 2 and Corollary 1 on bias–variance rates, Theorem 1 on measure conjugacy, and the new divergence–exponent bridge) require concrete simulation studies to validate the framework. The goal is to balance clarity with scope: provide simple, transparent evidence in this paper, while reserving extended stress tests and elaborations for a companion paper.

C.2.2 Core Simulations (to include in this manuscript)

1. Tier 1: One–dimensional controlled dynamics.

- System: autoregressive or scalar Markov process with conditional mean $\mu(y) = \text{sign}(y)|y|^q + \eta_t$.
- Purpose: verify $\eta = 2s$ from locality curves and $\zeta = \frac{2s}{2s+d}$ from learning curves with $d = 1$.
- Deliverables:
 - Log–log plot of $A(h) - A_0$ vs. h (bias slope η).
 - Learning curve $\min_h A_n(h) - A_0$ vs. n (rate ζ).
 - Table: recovered $(\hat{s}, \hat{d})$ compared to ground truth (s, d) .

2. Tier 3: Nonlinear 2D system with delay embedding.

- System: Hénon map with small noise; observation $Z_t = x_t$, embedded as $Y_t = [Z_t, Z_{t-1}, \dots, Z_{t-k+1}]$.
- Purpose: demonstrate Takens embedding + Rose equality $\Rightarrow$ measure conjugacy.
- Deliverables:
 - Divergence (MMD–JS or JS) between empirical edge law K_Y and pushforward $h_{\#}K_X$; report near-zero values (conjugacy test).
 - Log–log locality and learning curves in embedded space, recovering effective d smaller than ambient k .
 - Witness heatmap for model misspecification (e.g. perturbed Hénon parameter).

3. Tier 6: Dynamical exponents link.

- *Entropy/Lyapunov*: Logistic or Hénon map, where true entropy rate and Lyapunov exponents are known. Purpose: check that divergence slopes align with known entropy growth and Lyapunov stretching.
- *Mixing/Spectral*: Simple AR(1) or OU process, where correlation decay is exponential. Purpose: verify divergence contraction mirrors the mixing exponent (spectral gap).
- Deliverables:
 - Divergence decay curves vs. t , compared to theoretical entropy/Lyapunov or mixing rates.
 - Highlight slower e-projection in tail-driven regimes (recurrence), versus slower m-projection in bulk-driven regimes.

C.2.3 Extended Simulations (reserve for follow-up paper or appendix)

1. **Tier 2: State-dependent noise (heteroskedasticity).** Witness maps highlight local variance pockets invisible to RMSE.
2. **Tier 4: Ornstein–Uhlenbeck with nonlinear drift.** Demonstrates path-action connection (Proposition 3), stability gains from quadratic action regularization.
3. **Tier 5: Intrinsic vs. ambient dimension.** Synthetic manifold in high-dimensional space; estimator recovers effective dimension $d_\star$ rather than ambient dimension.

C.2.4 Summary

- *This paper:* Tiers 1, 3, and 6 — minimal but sufficient demonstrations linking theoretical rates, conjugacy, and dynamical exponents to concrete simulations.
- *Companion work:* Tiers 2, 4, 5 — richer stress tests and methodological elaborations, aimed at an applied ML or dynamical systems audience.